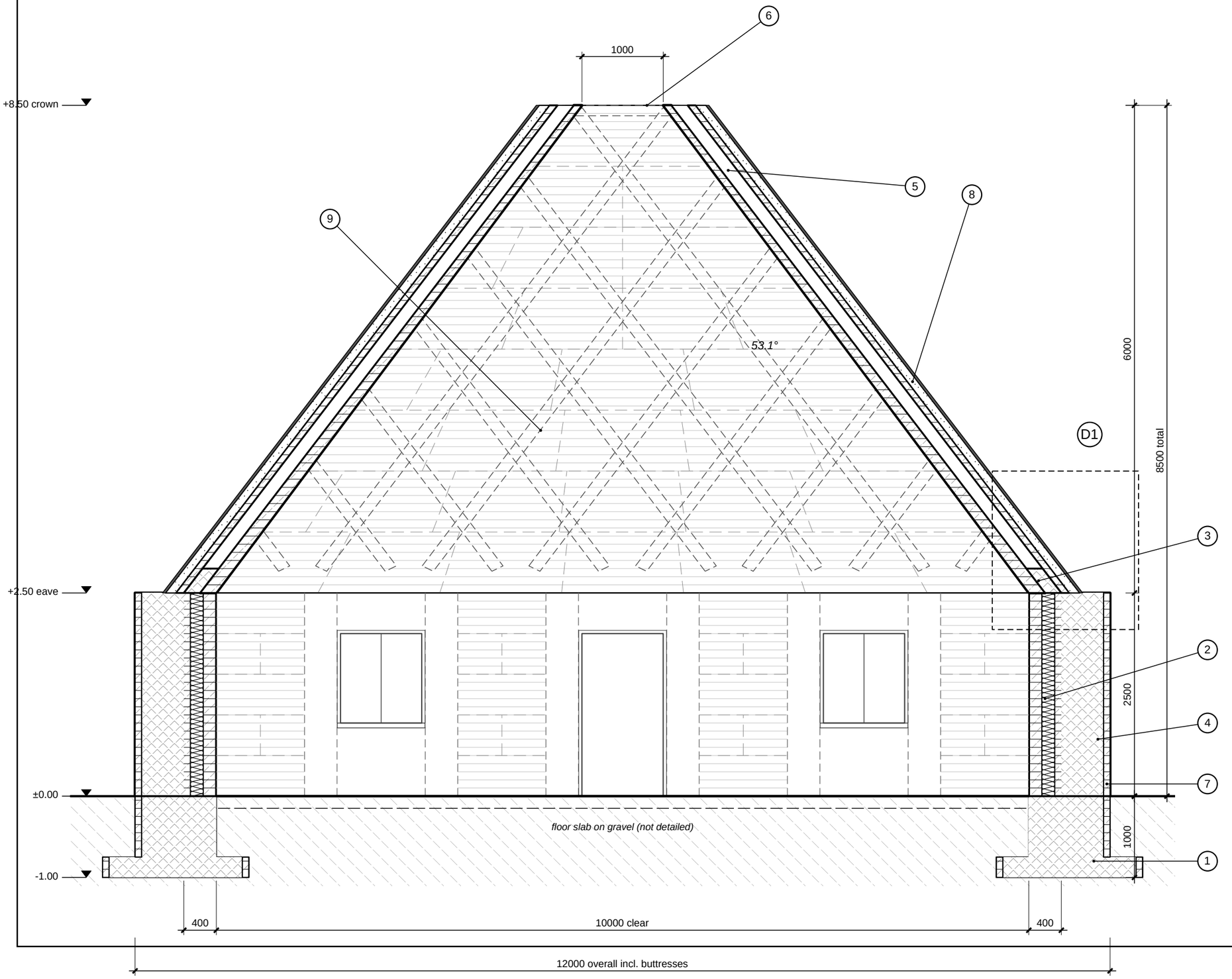
Total section A-A · grid B

through the east and west buttresses, looking north



1. Footing 1800 x 250, underside at -1000 (frost depth and bearing are guesses): printed edge formwork, cast core.
2. Printed wall 400, same pattern as the roof: two 80 inner layers, a 160 air zone, an 80 outer layer.
3. Solid base of the roof: over the first 300 (vertical) above the eave the 160 air zone is filled with cast concrete between the printed layers, so the 400 is solid; there is no ring beam.
4. Buttress 400 wide, 1000 from the inner face (600 beyond the wall), one about every 1.5 m (8 per side); printed formwork with cast core, rises to +2.50.
5. Roof shell: 400 thick square to the slope at the base (two 80 inner layers, a 160 air zone, an 80 outer layer), 320 at the crown; about 500 / 400 horizontally.
6. Crown opening 1000 clear; roof rises 6000 over 4500, slope 53.1° (36.9° from the vertical).
7. Everything is 3D printed: mortar beads in 10 mm layers (joints drawn every 100) form permanent formwork. Where a member is solid (roof base, buttress, footing, ribs) the printed shell is filled with cast concrete, crossed hatch. Single hatch = printed only.
8. Outside the printed shell: insulation 100 (dotted), then the outer layer (cladding 30, dark).
9. Cast ribs of the roof grid, seen on the back face as dashed outlines (roof grid B, diamond lattice). The cut lies in the air zone between ribs; rib sections are on the roof-grid sheets.

● from your sketches
○ assumed by me, to confirm

Notes

Layer pattern from your drawing; at the crown the second inner layer ends, so the leaf is 80 there (assumed).

The cut runs through the 4th buttress from the south (y = -743). Beyond it the north wall is seen from inside: every other bay is printed wall, the bays in between are openings (door in the middle, windows either side); buttresses behind the wall are dashed.

Faint lines = printed layer joints (every 100 drawn, real 10); dashed lines on the back face and the wall = divisions between printed pieces in running bond (assumed about 1500 wide).

0 1 2 3 4 5 m

Scale 1:50 at A3 · dimensions in mm · draft 1, 2026-10-11
pyramid-roof · hand estimates, not engineered